Lab: Envelope Modulation of Random Signals

Communications Systems Laboratory

Objective: Study how a continuous-time white Gaussian noise process is transformed through an Amplitude Modulation (AM) communication system:

1. *Baseband filtering*: RC low-pass filter with cutoff frequency f_n to create an *effectively low-pass* message signal
2. *Conventional AM modulation*: Multiplication by a carrier $[A_0 + m(t)]\cos(2\pi f_0 t)$ with $f_0 = kf_n$, $k = 30$
3. *Envelope detection*: Nonlinear demodulation using rectification and low-pass filtering

Compare the demodulated output with the original message signal to evaluate system performance. At each stage, analyze the signal characteristics by estimating *autocorrelation functions (ACFs)* and *power spectral densities (PSDs)*.

Theoretical Background

Conventional AM Signal. The conventional AM signal is given by:

$$s_{\text{AM}}(t) = (A_0 + m(t))\cos(2\pi f_0 t) \tag{1}$$

where A_0 is the carrier amplitude (constant), $m(t)$ is the message signal (baseband), and f_0 is the carrier frequency.

Modulation Index. For a random message process, the peak modulation index is interpreted over the *observed realization*:

$$\mu_{\text{peak}} = \frac{\max_t |m(t)|}{A_0} \tag{2}$$

Requirement: $\mu_{\text{peak}} < 1$ over the observation interval to avoid *overmodulation* (which causes envelope distortion). It is also useful to define the RMS modulation index

$$\mu_{\text{rms}} = \frac{\sigma_m}{A_0}. \tag{3}$$

Autocorrelation of the AM Signal. Let $m(t)$ be a zero-mean WSS random process with autocorrelation $R_{mm}(\tau)$. Since multiplication by a deterministic carrier produces a cyclostationary process, the following expression should be interpreted as the *cycle-averaged* autocorrelation (equivalently, it is exact if the carrier phase is assumed random and uniformly distributed):

$$\overline{R}_{s_{\text{AM}}}(\tau) = \frac{A_0^2}{2}\cos(2\pi f_0 \tau) + \frac{1}{2}R_{mm}(\tau)\cos(2\pi f_0 \tau) \tag{4}$$

Observation: The ACF exhibits oscillations at the carrier frequency f_0, modulated by the message ACF.

Power Spectral Density of the AM Signal. Under the same cycle-averaged/random-phase interpretation,

$$S_{s_{\text{AM}}}(f) = \frac{A_0^2}{4}(\delta(f - f_0) + \delta(f + f_0)) + \frac{1}{4}(S_{mm}(f - f_0) + S_{mm}(f + f_0)) \tag{5}$$

where $m(t)$ is assumed zero mean and WSS.

Envelope Detection. For proper AM with $\mu_{\text{peak}} < 1$ over the observed interval,

$$|A_0 + m(t)| = A_0 + m(t), \tag{6}$$

so the signal envelope equals

$$\text{Envelope}\{s_{\text{AM}}(t)\} = A_0 + m(t). \tag{7}$$

In this lab, the detector is implemented by *full-wave rectification* followed by low-pass filtering. This is an approximation to ideal envelope detection and may introduce a constant scale factor and residual ripple. After DC removal and normalization, the recovered signal should satisfy

$$\hat{m}(t) \approx m(t). \tag{8}$$

The implemented envelope detector consists of:

1. *Rectifier*: Converts the AM waveform into a nonnegative waveform
2. *Low-pass filter*: Smooths the rectified signal, removing the dominant high-frequency ripple generated by rectification

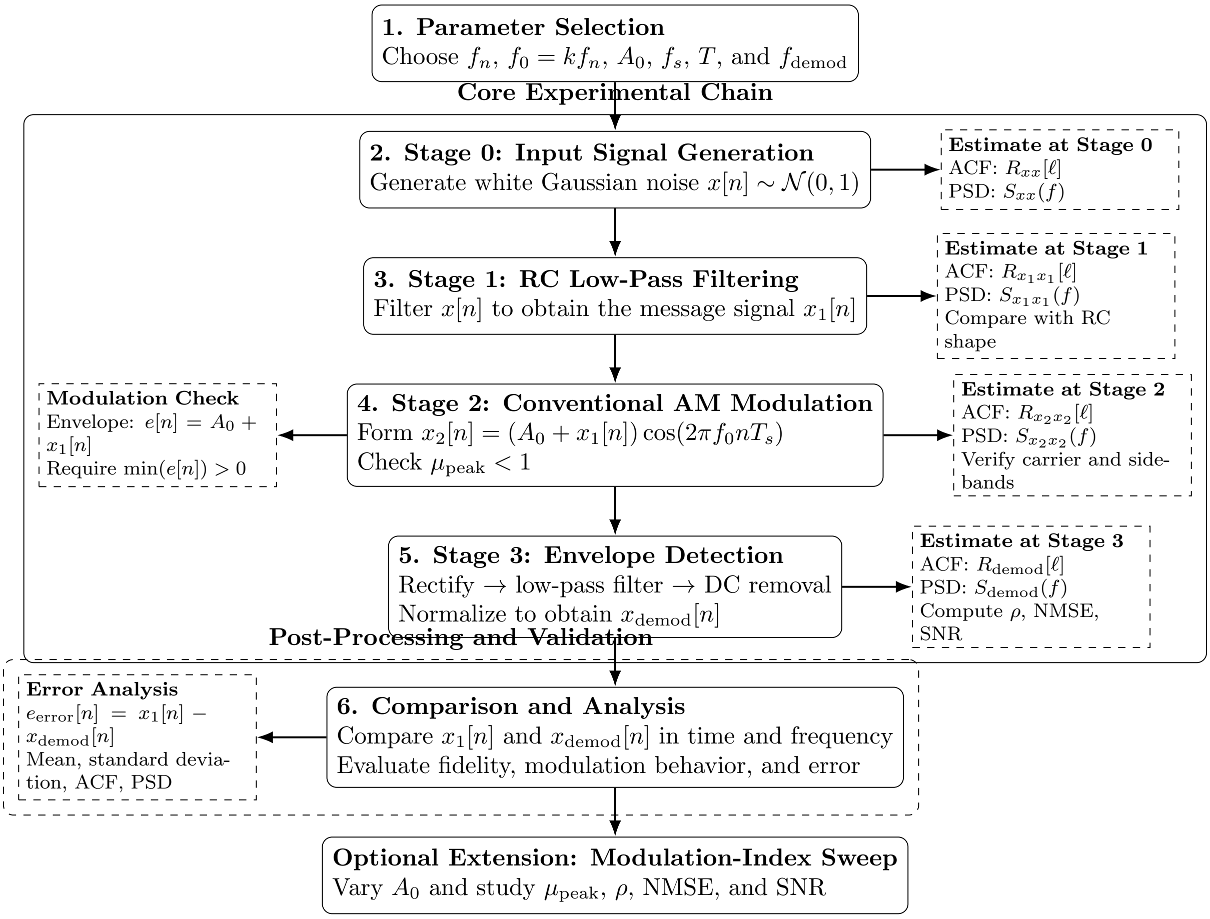

Figure 1: Flowchart of the lab procedure for envelope modulation of random signals.

Lab Procedure 1. Parameter Selection

Message Signal Parameters:

- Bandwidth parameter: $f_n = 10$ Hz
- RC components: $R_0 = 10$ kΩ, $C_0 = \frac{1}{2\pi f_n R_0} \approx 1.59$ μF

Carrier Parameters:

- Frequency: $f_0 = k \cdot f_n$ where $k = 30 \Rightarrow f_0 = 300$ Hz
- Amplitude: $A_0 = 4.0$ (assumes $\sigma_{x_1} \approx 1$ after normalization)
- Expected RMS modulation index: $\mu_{\mathrm{rms}} \approx 0.25$

Sampling Parameters:

- Sampling frequency: $f_s = 50 \cdot f_0 = 15{,}000$ Hz
- Rationale: $f_s \gg 2f_0 = 600$ Hz (Nyquist requirement with margin)
- Observation time: $T = \frac{500}{f_n} = 50$ seconds
- Number of samples: $N = f_s \cdot T = 750{,}000$
- Frequency resolution: $\Delta f = \frac{1}{T} = 0.02$ Hz $\ll f_n$

Demodulator Parameters:

- Cutoff frequency: $f_{\mathrm{demod}} = f_n = 10$ Hz
- Rationale: Preserves the message band while strongly attenuating the carrier-related high-frequency components
- Reference RC values for the equivalent cutoff: $R_1 = 10$ kΩ, $C_1 = \frac{1}{2\pi f_{\mathrm{demod}} R_1} \approx 1.59$ μF

PSD Estimation Parameters:

- Method: Welch's periodogram
- Segment length: $N_{\text{seg}} = 2^{14} = 16{,}384$ samples
- Overlap: 50% ($N_{\text{overlap}} = 8{,}192$ samples)
- Window: Hann window
- Frequency resolution: $\Delta f_{\text{Welch}} \approx \frac{f_s}{N_{\text{seg}}} \approx 0.92$ Hz

2. Stage 0: Generation of Input Signal
Procedure:

- Generate discrete-time white Gaussian noise $x[n]$ with zero mean and unit variance: $x[n] \sim \mathcal{N}(0, 1)$
- Estimate the autocorrelation function $R_{xx}[\ell]$ using the sample autocorrelation
- Estimate the power spectral density $S_{xx}(f)$ using Welch's method
- Plot the ACF and PSD of $x[n]$

Expected Results:

- **ACF**: $R_{xx}[\ell] \approx \delta[\ell]$ (impulse at zero lag, approximately zero elsewhere)
- **PSD**: $S_{xx}(f) \approx$ constant (flat spectrum across all frequencies)

3. Stage 1: RC Low-Pass Filtering
Procedure:

- Design a first-order RC low-pass filter with cutoff frequency $f_n = 10$ Hz
- Implement as a digital IIR filter (e.g., first-order Butterworth)
- Filter the white noise $x[n]$ to obtain the low-pass message signal $x_1[n]$
- Normalize $x_1[n]$ to unit variance
- Estimate $R_{x_1x_1}[\ell]$ and $S_{x_1x_1}(f)$
- Compare with the corresponding continuous-time RC shapes:

$$R_{x_1x_1}(\tau) \propto \exp(-2\pi f_n |\tau|) \tag{9}$$

$$S_{x_1x_1}(f) \propto \frac{1}{1 + (f/f_n)^2} \tag{10}$$

- Plot the ACF and PSD of $x_1[n]$ with theoretical overlays

Remark: Because the implementation is discrete-time and the output is normalized, the comparison with the RC formulas is approximate and should be interpreted mainly in terms of *shape*, not absolute scaling.
Expected Results:

- **ACF**: Exponential-like decay with time constant $\tau_c \approx 1/(2\pi f_n)$
- **PSD**: Low-pass characteristic with 3-dB corner approximately near $f_n = 10$ Hz

4. Stage 2: Conventional AM Modulation
Procedure:

- Set carrier parameters: $f_0 = 300$ Hz, $A_0 = 4.0$
- Generate carrier signal: $c[n] = \cos(2\pi f_0 n T_s)$ where $T_s = 1/f_s$
- Form conventional AM signal: $x_2[n] = (A_0 + x_1[n]) \cdot c[n]$
- Verify modulation index over the observation record:

$$\mu_{\text{peak}} = \frac{\max |x_1[n]|}{A_0} \tag{11}$$

- Ensure $\mu_{\text{peak}} < 1$ to avoid overmodulation
- Estimate $R_{x_2 x_2}[\ell]$ and $S_{x_2 x_2}(f)$
- Compare with the cycle-averaged theoretical ACF:

$$\overline{R}_{x_2 x_2}(\tau) \approx \left(\frac{A_0^2}{2} + \frac{1}{2} R_{x_1 x_1}(\tau) \right) \cos(2\pi f_0 \tau) \tag{12}$$

- Plot time-domain waveform (zoomed to show envelope)
- Plot ACF showing oscillations at f_0
- Plot PSD showing carrier and translated sidebands

Expected Results:

- **Time domain**: Carrier oscillations with amplitude varying according to $A_0 + x_1[n]$
- **ACF**: Oscillates at carrier frequency $f_0 = 300$ Hz
- **PSD**:
  - Discrete carrier lines at $\pm f_0$
  - Upper and lower sidebands centered around $\pm f_0$
  - Effective occupied bandwidth approximately $2f_n = 20$ Hz

5. Stage 3: Envelope Detection
Procedure:

- *Step 1 – Rectification*: Apply full-wave rectification: $x_{\text{rect}}[n] = |x_2[n]|$
- *Step 2 – Design demodulator filter*: Design low-pass filter with cutoff $f_{\text{demod}} = 10$ Hz (4th-order Butterworth recommended)
- *Step 3 – Filtering*: Filter the rectified signal: $x_{\text{env}}[n] = \text{LPF}\{x_{\text{rect}}[n]\}$
- *Step 4 – DC removal*: Remove DC component: $x_{\text{demod}}[n] = x_{\text{env}}[n] - \overline{x_{\text{env}}}$
- *Step 5 – Normalization*: Scale for comparison:

$$x_{\text{demod}}[n] \leftarrow x_{\text{demod}}[n] \cdot \frac{\sigma_{x_1}}{\sigma_{x_{\text{demod}}}} \tag{13}$$

- Estimate $R_{\text{demod}}[\ell]$ and $S_{\text{demod}}(f)$
- Compute performance metrics:

$$\rho = \text{corr}(x_1, x_{\text{demod}}) \tag{14}$$

$$\text{NMSE} = \frac{\overline{(x_1 - x_{\text{demod}})^2}}{\sigma_{x_1}^2} \tag{15}$$

$$\text{SNR} = -10 \log_{10}(\text{NMSE}) \text{ dB} \tag{16}$$

- Plot rectified signal, filtered envelope, and demodulated output
- Compare $x_1[n]$ and $x_{\text{demod}}[n]$ in time and frequency domains

Expected Results:

- **Correlation**: $\rho \approx 0.95$ to 0.99 (high fidelity)
- **ACF**: $R_{\text{demod}}(\tau)$ closely matches $R_{x_1 x_1}(\tau)$
- **PSD**: $S_{\text{demod}}(f)$ closely matches $S_{x_1 x_1}(f)$ over most of the message band
- **SNR**: > 20 dB for good demodulation quality

6. Comparison and Analysis
Baseband Signal Comparison:

- Plot $R_{x_1 x_1}(\tau)$ vs. $R_{\text{demod}}(\tau)$ on same axes

- Plot $S_{x_1x_1}(f)$ vs. $S_{\text{demod}}(f)$ on same axes
- Discuss demodulation fidelity based on correlation and NMSE

Modulation Effect Analysis:

- Compare $R_{x_1x_1}(\tau)$ (baseband) with $\overline{R}_{x_2x_2}(\tau)$ (modulated, cycle-averaged)
- Verify the theoretical relation for the modulated ACF
- Compare $S_{x_1x_1}(f)$ with $S_{x_2x_2}(f)$
- Verify spectral translation to $\pm f_0$

Modulation Index Verification:

- Compute peak modulation index: $\mu_{\text{peak}} = \max|x_1[n]|/A_0$
- Compute RMS modulation index: $\mu_{\text{rms}} = \sigma_{x_1}/A_0$
- Analyze envelope: $e[n] = A_0 + x_1[n]$
- Check for overmodulation: verify $\min(e[n]) > 0$
- Plot histograms of message signal and envelope
- Compute power efficiency:

$$\eta = \frac{P_{\text{sideband}}}{P_{\text{total}}} = \frac{\mu_{\text{rms}}^2}{1 + \mu_{\text{rms}}^2} \times 100\% \tag{17}$$

Error Analysis:

- Compute error signal: $e_{\text{error}}[n] = x_1[n] - x_{\text{demod}}[n]$
- Estimate error statistics: mean, standard deviation, maximum
- Compute and plot error ACF: $R_{\text{error}}(\tau)$
- Compute and plot error PSD: $S_{\text{error}}(f)$
- Discuss sources of demodulation error:
    1. Rectifier nonlinearity (harmonics and intermodulation)
    2. Low-pass filter limitations (finite rolloff, carrier leakage)
    3. Statistical estimation errors (finite observation time)
    4. Numerical precision (floating-point roundoff)
    5. Near-overmodulation effects (if μ_{peak} is close to 1)

Optional Extension: Modulation Index Sweep *Objective*: Study the effect of varying modulation index on demodulation performance.

Procedure:

1. Vary carrier amplitude: $A_0 \in [2.0, 6.0]$ (9 values)
2. For each A_0:
    - Modulate: $x_2 = (A_0 + x_1) \cdot c$
    - Check for overmodulation: $\min(A_0 + x_1) < 0$?
    - Demodulate using the envelope detector
    - Compute: μ_{peak}, ρ, NMSE, SNR
3. Plot: ρ vs. μ_{peak}, NMSE vs. μ_{peak}, SNR vs. μ_{peak}
4. Identify an optimal A_0 as a trade-off between distortion avoidance and power efficiency

Expected Observations:

- For $\mu_{\text{peak}} < 1$: High correlation ($\rho > 0.95$), low distortion
- For $\mu_{\text{peak}} \approx 1$: Slight degradation as the envelope approaches zero
- For $\mu_{\text{peak}} > 1$: Severe distortion due to overmodulation

Required Deliverables Submit a lab report containing:

Plots:

1. Stage 0: ACF and PSD of white noise
2. Stage 1: ACF and PSD of filtered message (with theoretical overlay)
3. Stage 2: Time-domain waveform showing envelope, ACF, and PSD
4. Stage 3: Envelope detection process and comparison plots
5. Modulation index analysis (4 subplots)
6. Error analysis (4 subplots)
7. Comprehensive comparison (2×2 grid)

Calculated Metrics:

1. Peak and RMS modulation indices (μ_{peak}, μ_{rms})
2. Correlation coefficient ρ between x_1 and x_{demod}
3. Normalized mean square error (NMSE)
4. Signal-to-noise ratio (SNR) in dB
5. Power efficiency η
6. Bandwidth utilization